

Name _____



THE
KING'S
SCHOOL

2020

Year 12
Higher School Certificate Course
Assessment Task 2 (12AT2)
March 2020

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time – 1.5 hours
- Write using **black pen**
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 50

Section I 5 marks

- Answer Questions 1-5 on the multiple choice answer sheet provided
- Allow about 10 minutes for this section

Section II 45 marks

- Answer Questions 6 -8 in the spaces provided
- Allow about 80 minutes for this section

Examiners' Use Only			Setter/Checker: GJD/BDB	
Question	Complex Numbers	Proof		Totals
Section I				
1-5	3,4, 5 / 3	1, 2 / 2	/ 5	
Section II				
6	/15			/15
7	a,b,c / 8	d,e / 7	/15	
8		/15	/15	
Total	/26	/24	/50	

Name

Section I

5 marks

Attempt Questions 1 - 5

Allow about 10 minutes for this section

1. Consider the statements:

P : n is a positive integer.

Q : n is an even number greater than 0.

Which of the following is false?

(A) $Q \Rightarrow P$

(B) $P \Leftarrow Q$

(C) P is implied by Q

(D) P implies Q

2. Suppose that both x and y are odd. Which of the following statements is false?

(A) $x + y$ is even

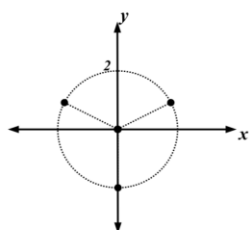
(B) $x - y$ is even

(C) $3x + 5y$ is even

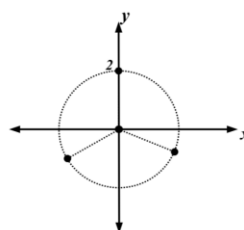
(D) xy is odd

3. Which complex plane best represents the cube roots of $8i$?

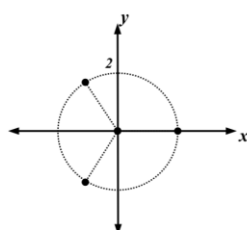
(A)



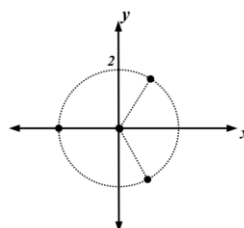
(B)



(C)



(D)



Name _____

4. The complex number z satisfies $|z - 2| = 2$. What is the greatest distance that z can be from the point i in the complex plane?

- (A) 1
(B) 2
(C) $\sqrt{5} + 2$
(D) $2\sqrt{2}$

5. For a certain complex number z where $\text{Arg}(z) = \frac{\pi}{5}$
 $\text{Arg}(z^7)$ is:

- (A) $\frac{-7\pi}{5}$
(B) $\frac{-3\pi}{5}$
(C) $\frac{2\pi}{5}$
(D) $\frac{3\pi}{5}$

Name

Section II

45 marks

Attempt Questions 6-8

Allow about 80 minutes for this section

All questions are of equal value

Question 6 (15 marks) START A NEW WRITING BOOKLET

Marks

- (a) Prove that $2 + i$ is a root of $z^3 - 3z^2 + z + 5 = 0$, and hence find the other roots. 3
- (b) (i) Write $\frac{\sqrt{3}+i}{\sqrt{3}-i}$, in the form $a + ib$ where a and b are real. 1
- (ii) By expressing $\sqrt{3} + i$ and $\sqrt{3} - i$ in polar form, write $\frac{\sqrt{3}+i}{\sqrt{3}-i}$ in polar form. 2
- (iii) Hence, find $\sin \frac{\pi}{3}$ in surd form. 1
- (c) (i) Find the square roots of $-6 + 8i$. 2
- (ii) Hence solve the equation $z^2 - \sqrt{2}iz + 1 - 2i = 0$. 2
- (d) Find the eighth roots of 1 and show these roots in the complex plane. 4

End of Question 6

Name _____

Question 7 (15 marks) START A NEW WRITING BOOKLET

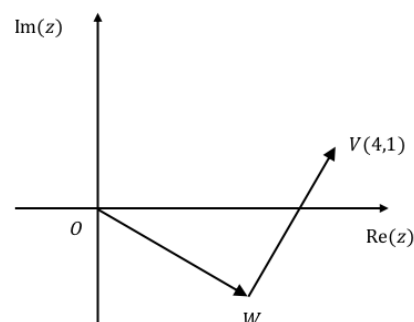
Marks

- (a) The point V represents the complex number $4 + i$.

3

$$\angle OWV = \frac{\pi}{2} \text{ and } |VW| = |OW|.$$

Find the complex number represented by W .



- (b) Prove that if ω is a non-real cube root of unity,

$$\text{then } (1 + \omega - \omega^2)^7 = -128\omega^2.$$

2

- (c) Identify subsets of the complex plane determined by the relation:

3

$$|z - 2i| \geq 4|z + 3|$$

- (d) Prove by Mathematical Induction that $4^{n+1} + 6^n$ is divisible by 10 when n is even.

3

- (e) Given $a_1 = 2$ and $a_n = \frac{a_{n-1}}{n}$, for $n \geq 2$,

prove, using Mathematical Induction, that $a_n = \frac{2}{n!}$ for $n \geq 1$.

4

End of Question 7

Name _____

Question 8 (15 marks) START A NEW WRITING BOOKLET

Marks

- (a) Find the negation of the following:

$\forall$ real numbers $x, x < 2m+1$ for integral values of m .

1

- (b) Prove by contrapositive that if $\frac{mn}{2}$ is integral then m or n must be even.

3

- (c) Prove $\sqrt{2} + \sqrt{3} < 10$ by contradiction.

3

- (d) Prove that a number is divisible by 4 if and only if the last two digits are a multiple of 4. Let the number be $x = 100a + 10b + c$ where a, b, c are positive integers and $b, c \leq 9$.

Do NOT use Proof by Induction.

4

- (e) Prove

$$\frac{x^2}{y^2} + \frac{y^2}{z^2} + \frac{z^2}{x^2} \geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4} + \frac{1}{xz} \sqrt{z^2 y^2 + x^4} \text{ for } x, y, z > 0.$$

4

End of assessment

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Mathematics Extension 2 Solutions

General Instructions

- Reading time - 5 minutes
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- A reference sheet is provided
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Total marks: 50

Section I 5 marks

- Answer Questions 1-5 on the multiple choice answer sheet provided
- Allow about 5 minutes for this section

Section II 45 marks

- Answer Questions 6 -8 in the spaces provided
- Allow about 85 minutes for this section

Examiners' Use Only			Setter/Checker: GJD/BJB	
Question	Complex Numbers	Proof	Totals	
Section I				
1-5	3,4, 5 / 3	1, 2 / 2	/5	
Section II				
6	/15		/15	
7	a,b,c /8	d,e /7	/15	
8		/15	/15	
Total	/26	/24	/50	

Name _____

Section I

5 marks

Attempt Questions 1 - 5

Allow about 10 minutes for this section

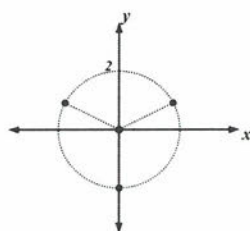
1. Consider the statements:

P : n is a positive integer.

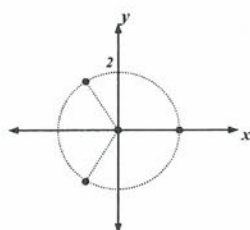
Q : n is an even number greater than 0.

Which of the following is false?

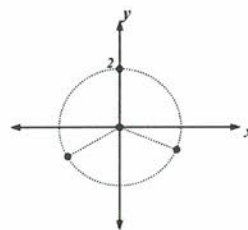
- (A) $Q \Rightarrow P$ True Since a number must be an integer if it is even
- (B) $P \Leftarrow Q$ True Since all even numbers are integers
- (C) P is implied by Q True Since all even numbers are integers
- (D) P implies Q False Since if a number is an integer it might not be even
2. Suppose that both x and y are odd. Which of the following statements is false?
- (A) $x + y$ is even
- (B) $x - y$ is even
- (C) $3x + 5y$ is even
- (D) xy is odd
3. Which complex plane best represents the cube roots of $8i$?



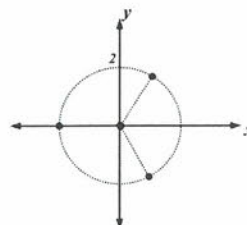
(C)



(B)



(D)



Name _____

4. The complex number z satisfies $|z - 2| = 2$. What is the greatest distance that z can be from the point i in the complex plane?
- (A) 1
- (B) 2
- (C) $\sqrt{5} + 2$
- (D) $2\sqrt{2}$
5. For a certain complex number z where $\text{Arg}(z) = \frac{\pi}{5}$
 $\text{Arg}(z^7)$ is:
- (A) $\frac{-7\pi}{5}$
- (B) $\frac{-3\pi}{5}$
- (C) $\frac{2\pi}{5}$
- (D) $\frac{3\pi}{5}$

Name _____

Section 2

45 marks

Attempt Questions 6-8

Allow about 80 minutes for this section

All questions are of equal value

Question 6 (15 marks) START A NEW WRITING BOOKLET

Marks

- (a) Prove that $2 + i$ is a root of $z^3 - 3z^2 + z + 5 = 0$, and hence find the other roots.

$$\begin{aligned}(2 + i)^3 - 3(2 + i)^2 + (2 + i) + 5 \\&= 2^3 + 3(2)^2i + 3(2)(i)^2 + i^3 - 3(2^2 + 4i + i^2) + (2 + i) + 5 \\&= 8 + 12i - 6 - i - 12 - 12i + 3 + 2 + i + 5 \\&= 0\end{aligned}$$

$\therefore 2 + i$ is a root.

$\therefore 2 - i$ is a root.

$$\begin{aligned}\therefore 2 + i + 2 - i + \gamma &= -\frac{b}{a} \\&= -\frac{-3}{1} \\&= 3\end{aligned}$$

$$\therefore 4 + \gamma = 3$$

$$\therefore \gamma = -1$$

$\therefore$ The roots of $z^3 - 3z^2 + z + 5 = 0$ are $-1, 2 \pm i$.

Name _____

- 6(b) (i) Write $\frac{\sqrt{3}+i}{\sqrt{3}-i}$, in the form $a + ib$ where a and b are real.

$$\begin{aligned}\frac{\sqrt{3}+i}{\sqrt{3}-i} &\times \frac{\sqrt{3}+i}{\sqrt{3}+i} \\&= \frac{3 + 2\sqrt{3}i + 1}{3 + 1} \\&= \frac{4 + 2\sqrt{3}i}{4} \\&= 1 + \frac{\sqrt{3}}{2}i\end{aligned}$$

- (ii) By expressing $\sqrt{3} + i$ and $\sqrt{3} - i$ in polar form, write $\frac{\sqrt{3}+i}{\sqrt{3}-i}$ in polar form.

$$\begin{aligned}\sqrt{3} + i &= 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \\ \sqrt{3} - i &= 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) \\ \frac{\sqrt{3} + i}{\sqrt{3} - i} &= \frac{2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)}{2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right)} \\&= \frac{2}{2} \left(\cos \left(\frac{\pi}{6} - -\frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{6} - -\frac{\pi}{6} \right) \right) \\&= \cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right)\end{aligned}$$

- (iii) Hence, find $\sin \frac{\pi}{3}$ in surd form.

Equating imaginary parts of **a** and **b**

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

Name _____

- 6(c) (i) Find the square roots of $-6 + 8i$.

$$\begin{aligned} |-6 + 8i| &= \sqrt{(-6)^2 + 8^2} = 10, \operatorname{Re}(-6 + 8i) \\ &= -6 \end{aligned}$$

The square roots of $-6 + 8i$ are

$$\pm \left(\sqrt{\frac{10 - 6}{2}} + \sqrt{\frac{10 + 6}{2}}i \right) = \pm(2 + 2\sqrt{2}i)$$

- (ii) Hence solve the equation $z^2 - \sqrt{2}iz + 1 - 2i = 0$.

$$z = \frac{\sqrt{2}i \pm \sqrt{(-\sqrt{2}i)^2 - 4(1)(1 - 2i)}}{2(1)}$$

$$= \frac{\sqrt{2}i \pm \sqrt{-2 - 4 + 8i}}{2}$$

$$= \frac{\sqrt{2}i \pm \sqrt{-6 + 8i}}{2}$$

$$= \frac{\sqrt{2}i \pm (2 + 2\sqrt{2}i)}{2}$$

$$= -2 - \sqrt{2}i, 2 + 3\sqrt{2}i$$

Name _____

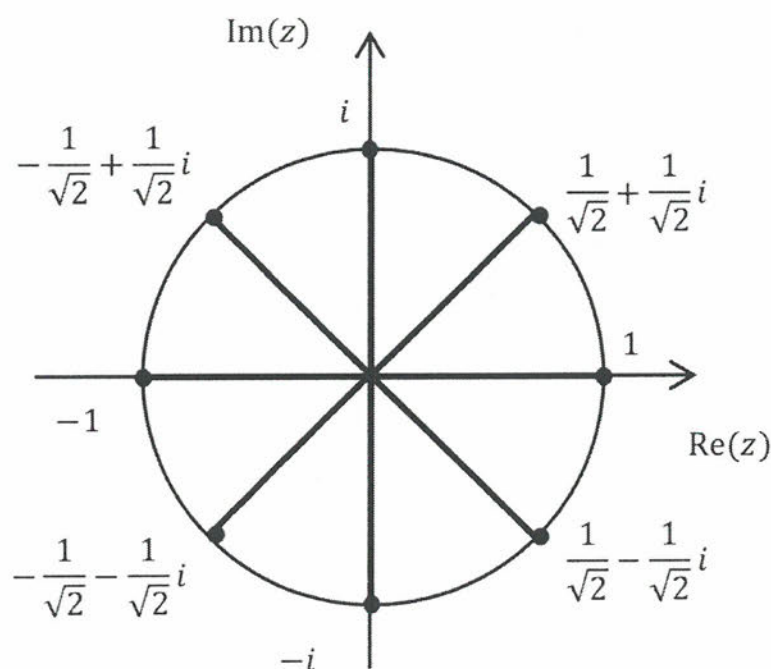
- 6(d) Find the eighth roots of 1 ($z^8 = 1$) in exponential form and show these roots in the complex plane.

4

1 is an eighth root of 1, and the other roots must be spaced by $\frac{2\pi}{8} = \frac{\pi}{4}$ as shown. The eighth roots of 1 are $\pm 1, \frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i, \pm i, -\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i$.

Alternatively in exponential form the roots are

$$1, e^{\pm \frac{\pi}{4}i}, \pm i, e^{\pm \frac{3\pi}{4}i}$$



End of Question 6

Name _____

Question 7 (15 marks) START A NEW WRITING BOOKLET

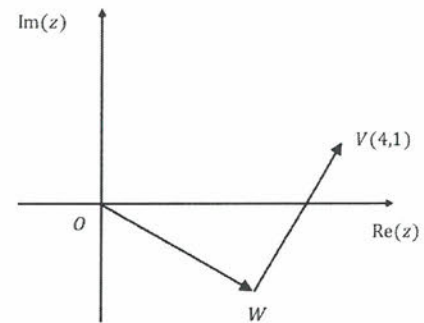
Marks

- (a) The point V represents the complex number $4 + i$.

3

$$\angle OWV = \frac{\pi}{2} \text{ and } |VW| = |OW|.$$

Find the complex number represented by W .



let W represent $a + ib$

$$-i\overrightarrow{WV} = \overrightarrow{OW}$$

$$-i(4 + i - a - ib) = a + ib$$

$$-4i + 1 + ai - b = a + bi$$

$$-b + 1 + i(a - 4) = a + bi$$

$$\therefore -b + 1 = a \quad b = 1 - a$$

$$a - 4 = b$$

$$a - 4 = 1 - a$$

$$2a = 5$$

$$a = \frac{5}{2}$$

$$b = 1 - \frac{5}{2}$$

$$= -\frac{3}{2}$$

W represents $\frac{5-3i}{2}$

Name _____

7(b) Prove that if ω is a non-real cube root of unity,

$$\text{then } (1 + \omega - \omega^2)^7 = -128\omega^2.$$

2

$$\begin{aligned} 1 + \omega + \omega^2 &= 0 \\ \therefore (1 + \omega - \omega^2)^7 &= (1 + \omega + \omega^2 - 2\omega^2)^7 \\ &= (-2\omega^2)^7 \\ &= -128\omega^{14} \\ &= -128(\omega^3)^4 \omega^2 \\ &= -128\omega^2 \end{aligned}$$

(c) Identify subsets of the complex plane determined by relation:

3

$$|z - 2i| \geq 4|z + 3|$$

Let $z = x + iy$.

$$\therefore |x + iy - 2i| \geq 4|x + iy + 3|$$

$$|x + (y - 2)i| \geq 4|(x + 3) + iy|$$

$$\sqrt{x^2 + (y - 2)^2} \geq 4\sqrt{(x + 3)^2 + y^2}$$

$$x^2 + (y - 2)^2 \geq 16((x + 3)^2 + y^2)$$

$$x^2 + y^2 - 4y + 4 \geq 16(x^2 + 6x + 9 + y^2)$$

$$x^2 + y^2 - 4y + 4 \geq 16x^2 + 96x + 144 + 16y^2$$

$$15x^2 + 96x + 15y^2 + 4y + 140 \leq 0$$

$$x^2 + \frac{96}{15}x + y^2 + \frac{4}{15}y + \frac{140}{15} \leq 0$$

$$x^2 + \frac{96}{15}x + \left(\frac{48}{15}\right)^2 + y^2 + \frac{4}{15}y + \left(\frac{2}{15}\right)^2 + \frac{140}{15} \leq \left(\frac{48}{15}\right)^2 + \left(\frac{2}{15}\right)^2$$

$$\left(x + \frac{48}{15}\right)^2 + \left(y + \frac{2}{15}\right)^2 \leq \frac{208}{225}$$

Name _____

7(d) Prove by Mathematical Induction that $4^{n+1} + 6^n$ is divisible by 10 when n is even.

3

Let $P(n)$ represent the proposition.

$P(2)$ is true since $4^{2+1} + 6^2 = 100 = 10(10)$

If $P(k)$ is true for some arbitrary $k \geq 2$ then $4^{k+1} + 6^k = 10m$ for integral m

RTP $P(k+2)$ $4^{k+3} + 6^{k+2} = 10p$ for integral p

$$\begin{aligned}\text{LHS} &= 16(4^{k+1}) + 36(6^k) \\ &= 16(4^{k+1} + 6^k) + 20(6^k) \\ &= 16(10m) + 20(6^k) \quad \text{from } P(k) \\ &= 10(16m + 2 \times 6^k) \\ &= 10p \quad \text{for integral } p \text{ since } m \text{ and } k \text{ are integral} \\ &= \text{RHS}\end{aligned}$$

$$\therefore P(k) \Rightarrow P(k+2)$$

$\therefore P(n)$ is true for even $n \geq 2$ by induction $\square$

Name _____

7(e) Given $a_1 = 2$ and $a_n = \frac{a_{n-1}}{n}$, for $n \geq 2$,
prove, using Mathematical Induction, that $a_n = \frac{2}{n!}$ for $n \geq 1$.

4

Let $P(n)$ represent the proposition.

$P(1)$ is true since $a_1 = 2$ and $a_1 = \frac{2}{1!} = 2$

If $P(k)$ is true for some arbitrary $k \geq 1$ then $a_k = \frac{2}{k!}$

RTP $P(k+1)$ $a_{k+1} = \frac{2}{(k+1)!}$

LHS = a_{k+1}

$= \frac{a_k}{k+1}$ from the recursive formula

$= \frac{2}{k!} \div (k+1)$ from $P(k)$

$= \frac{2}{(k+1)!}$

= RHS

$\therefore P(k) \Rightarrow P(k+1)$

$\therefore P(n)$ is true for $n \geq 1$ by induction $\square$

Name _____

Question 8 (15 marks) START A NEW WRITING BOOKLET

Marks

- (a) Find the negation of the following:

$\forall$ real numbers $x, x < 2m + 1$ for integral m .

$\exists$ real numbers $x, x \geq 2m + 1$ for integral m

- (b) Prove by contrapositive that if $\frac{mn}{2}$ is integral then m or n must be even.

Suppose m is odd and n is odd.

Let $m = 2k + 1, n = 2j + 1$ for integral k, j

$$\begin{aligned}\frac{mn}{2} &= \frac{(2k + 1)(2j + 1)}{2} \\ &= \frac{4kj + 2k + 2j + 1}{2} \\ &= 2kj + k + j + \frac{1}{2}\end{aligned}$$

Which is not integral since j and k are integral

$\therefore$ if m and n are both odd then $\frac{mn}{2}$ is not integral.

$\therefore$ if $\frac{mn}{2}$ is integral then m or n must be even.

- (c) Prove $\sqrt{2} + \sqrt{3} < 10$ by contradiction.

Suppose $\sqrt{2} + \sqrt{3} \geq \sqrt{10}$

*

$$(\sqrt{2} + \sqrt{3})^2 \geq 10 \quad \text{since } \sqrt{2}, \sqrt{3}, \sqrt{10} > 0$$

$$2 + 2\sqrt{6} + 3 \geq 10$$

$$2\sqrt{6} \geq 5$$

$$\sqrt{24} \geq \sqrt{25}$$

$$24 \geq 25$$

#

Which is a contradiction, so $\sqrt{2} + \sqrt{3} < \sqrt{10}$

Name _____

- 8(d) Prove that a number is divisible by 4 if and only if the last two digits are a multiple of 4.
Let the number be $x = 100a + 10b + c$ where a, b, c are positive integers and $b, c \leq 9$. Do not use Proof by Induction.

let the number be $x = 100a + 10b + c$ where a, b, c are positive integers and $b, c \leq 9$

If the last two digits are a multiple of 4 then $10b + c = 4m$ for integral m

$$\therefore x = 4(25a) + 4m$$

$$= 4(25a + m)$$

$$= 4p \text{ for integral } p \text{ since } a, m \text{ are integral}$$

$\therefore$ if the last two digits are a multiple of 4 then the number is divisible by 4.

Conversely, we will show by contrapositive that if a number is divisible by 4 then the last two digits are a multiple of 4.

If the last two digits are not a multiple of 4 then $10b + c = 4m + k$ for integral m, k with k not a multiple of 4.

$$\therefore x = 4(25a) + 4m + k$$

$$= 4(25a + m) + k$$

$$\neq 4p \text{ for integral } p \text{ since } a, m \text{ are integral and } k \text{ is not a multiple of 4}$$

$\therefore$ if the last two digits are not a multiple of 4 then the number is not divisible by 4.

$\therefore$ if the number is divisible by 4 then the last two digits are a multiple of 4

$\therefore$ a number is divisible by 4 if and only if the last two digits are a multiple of 4.

Name _____

8(e) Prove $\frac{x^2}{y^2} + \frac{y^2}{z^2} + \frac{z^2}{x^2} \geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4} + \frac{1}{xz} \sqrt{z^2 y^2 + x^4}$ for $x, y, z > 0$.

4

$$\frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} \geq \sqrt{\frac{x^2}{y^2} \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)} \quad (\text{AM} - \text{GM})$$

$$\frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} \geq \sqrt{\frac{x^2}{y^2} \left(\frac{x^2 y^2 + z^4}{x^2 z^2}\right)}$$

$$\frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} \geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4} \quad (1)$$

Similarly

$$\frac{\frac{y^2}{z^2} + \left(\frac{x^2}{y^2} + \frac{z^2}{x^2}\right)}{2} \geq \frac{1}{xz} \sqrt{y^2 z^2 + x^4} \quad (2)$$

(1) + (2):

$$\frac{x^2}{y^2} + \frac{y^2}{z^2} + \frac{z^2}{x^2} \geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4} + \frac{1}{xz} \sqrt{y^2 z^2 + x^4}$$

End of Assessment